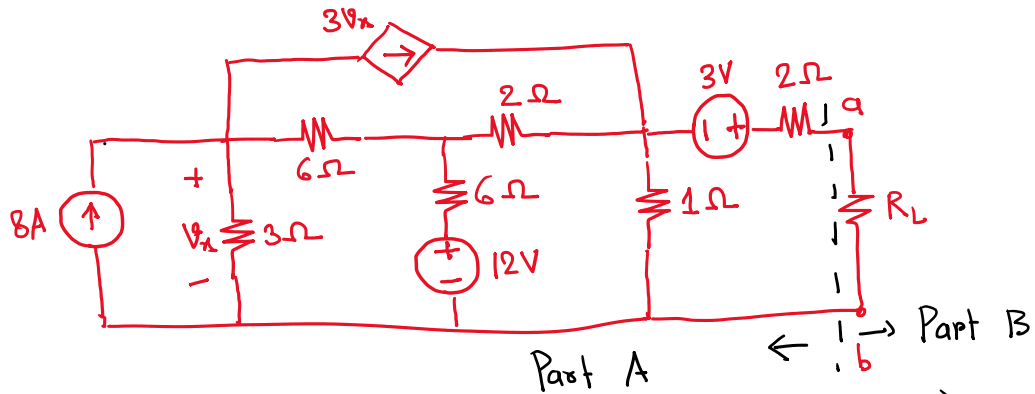
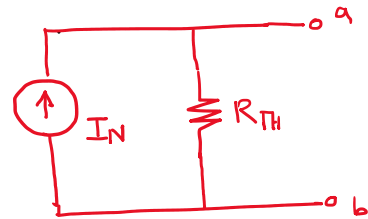
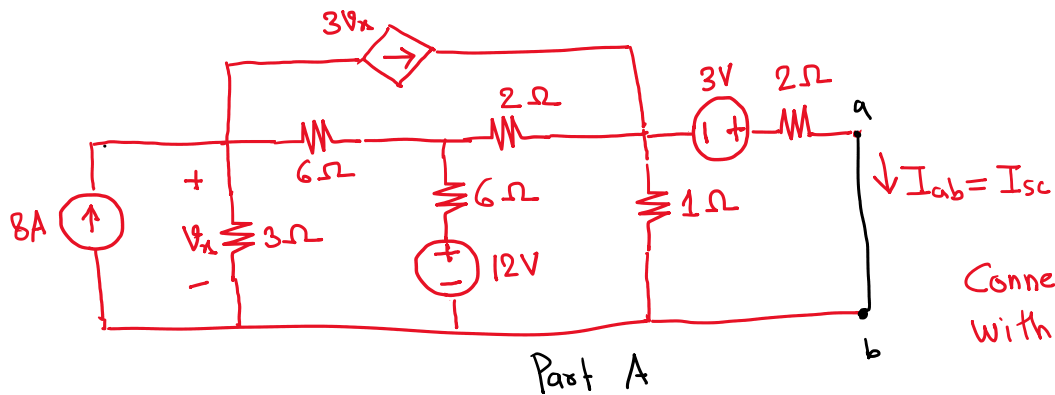


Norton's Theorem:

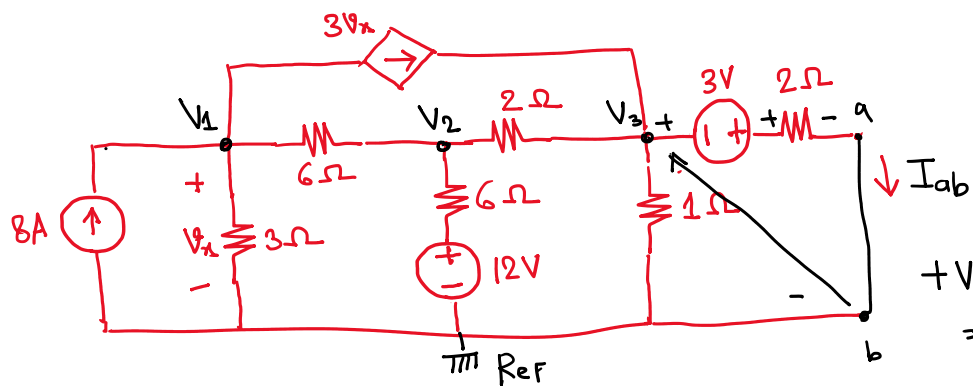


Remove Part B. (only Part A Remaining)



Connect a & b with short ckt.

$$I_N = I_{sc} = I_{ab}$$



$$+V_3 + 3 - 2I_{ab} = 0$$

$$\Rightarrow I_{ab} = \frac{V_3 + 3}{2}$$

Node 1: $\frac{V_1}{2} + \frac{V_1 - V_2}{6} - 8 + 3V_x = 0$

$V_x = V_1 - 0 = V_1$

$$\text{Node 1: } \frac{V_1}{3} + \frac{V_1 - V_2}{6} - 8 + 3V_x = 0 \quad V_x = V_1 - 0 = V_1$$

$$\Rightarrow \frac{V_1}{3} + \frac{V_1 - V_2}{6} - 8 + 3V_1 = 0 \quad \text{--- (i)}$$

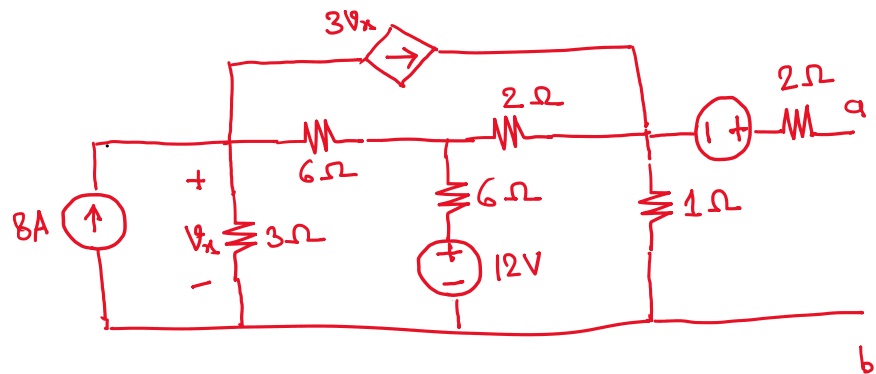
$$\text{Node 2: } \frac{V_2 - V_1}{6} + \frac{V_2 - V_3}{2} + \frac{V_2 - 12 - 0}{6} = 0 \quad \text{--- (ii)}$$

$$\text{Node 3: } \frac{V_3 - 0}{1} + \frac{V_3 - V_2}{2} - 3V_x + \frac{V_3 + 3 - 0}{2} = 0$$

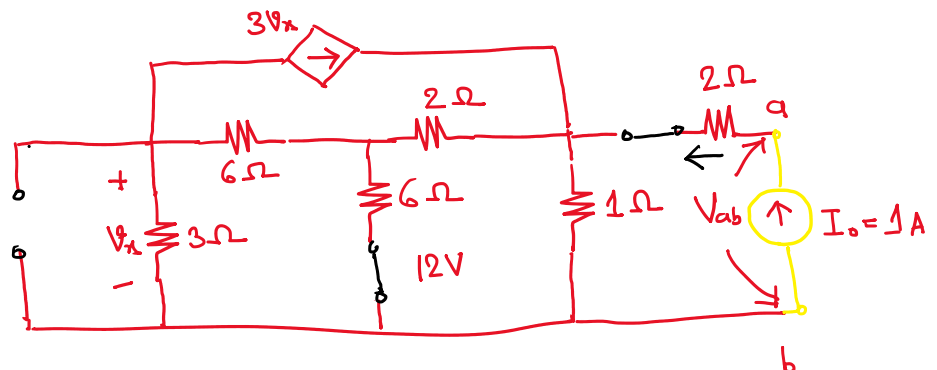
$$\frac{V_3 - 0}{1} + \frac{V_3 - V_2}{2} - 3V_1 + \frac{V_3 + 3}{2} = 0 \quad \text{--- (iii)}$$

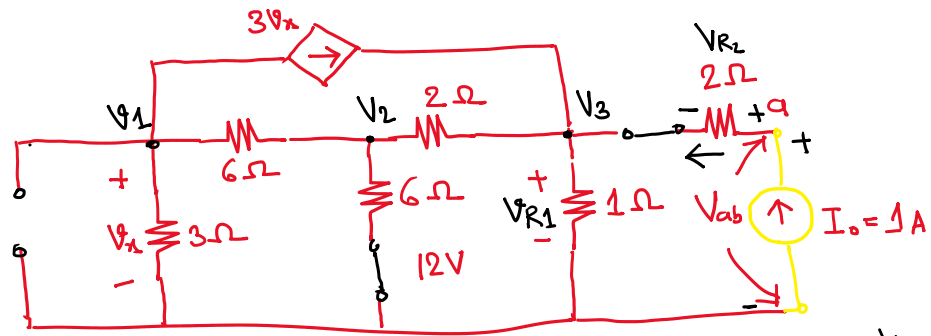
V_1, V_2, V_3 [After solving the equations]

$$I_N = I_{sc} = I_{ab} = \frac{V_3 + 3}{2}$$



To determine R_{Th} : 1. Remove Independent Sources
 2. IF dependent Source exists, place an arbitrary Current Source (say $I_o = 1A$) between the nodes a & b



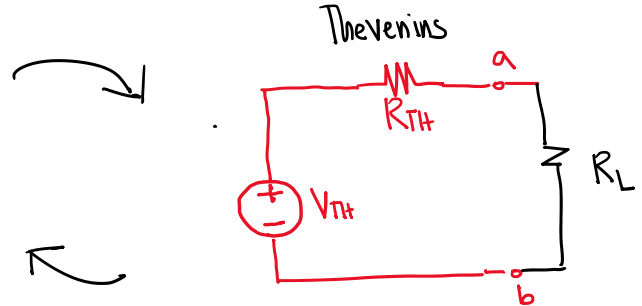
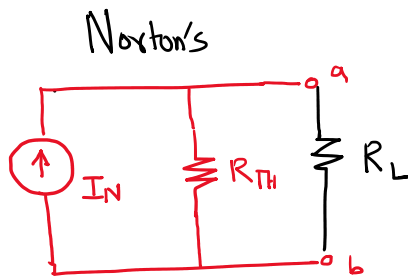


Apply Node Analysis method to determine V_3 $+V_{R1}+V_{R2}-V_{ab}=0$
 $\Rightarrow V_{ab} = V_{R1} + V_{R2}$

$$\frac{V_3 - V_2}{2} + \frac{V_3}{4} - 3V_x + I_o = 0$$

$$\Rightarrow V_3 = V_{R1}$$

$$R_{TH} = \frac{V_{ab}}{I_o}$$



Applying Source transformation

$$V_{TH} = I_N R_{TH}$$

$$I_N = \frac{V_{TH}}{R_{TH}}$$